ALICE L WALTON SCHOOL OF MEDICINE

Intestinal Neoplasms

Last updated July 15, 2026 · 21 min read · ScholarRx

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Adapted from: Intestinal Neoplasms

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Compare and contrast the histologic types of intestinal polyps, including their genetic associations and clinical significance.
2. Describe the genetics and malignancies associated with the included hereditary colon cancer syndromes.
3. Differentiate the clinical presentation, pathophysiology, diagnosis, and manage oid tumor.

CASE CONNECTION

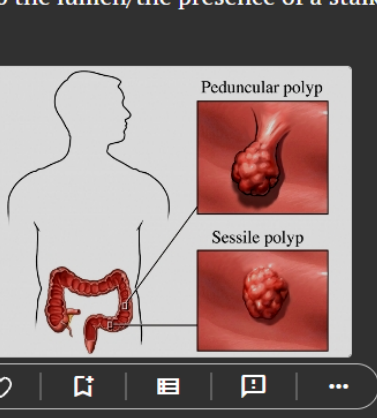
A 52-year-old asymptomatic patient, SC, with a family history of colon cancer is due for screening colonoscopy.

How will you explain to SC why you think a screening colonoscopy is important? Consider your answer as you read, and we'll revisit SC at the end of the brick.

GO TO CONCLUSION ↓

What Is an Intestinal Polyp (LO #1)?

An intestinal polyp is defined as a mucosal protuberance into the lumen. Grossly, they are characterized as flat, sessile, or pedunculated based on the degree of protrusion into the lumen/the presence of a stalk (Figure 1).



CLINICAL CORRELATION

Polyps are most common in the colon and rectum but can also occur in the esophagus, stomach, or small intestine.

In general, intestinal polyps are classified as non-neoplastic and neoplastic. Non-neoplastic refers to non-cancerous tissue changes (eg, inflammatory, degenerative, adaptive, or reparative). The most common neoplastic polyps are adenomatous and serrated polyps, which have the potential to progress to cancer.

Most polyps are asymptomatic but may cause bleeding, pain with defecation, tenesmus (rectal), or obstruction if large.

Non-Neoplastic Polyps

Non-neoplastic polyps are classified as inflammatory, hamartomatous, or hyperplastic.

Hyperplastic Polyps

Colonic hyperplastic polyps are benign epithelial proliferations that are usually discovered in the 6th-7th decades of life (often incidentally found during colonoscopy). They are the most common polyps, lack malignant potential, and must be distinguishe

Hyperplastic polyps are most commonly found in the left colon and usually <5 mm in diameter. They are composed of mature goblet and absorptive cells with a serrated surface architecture (Figure 2).

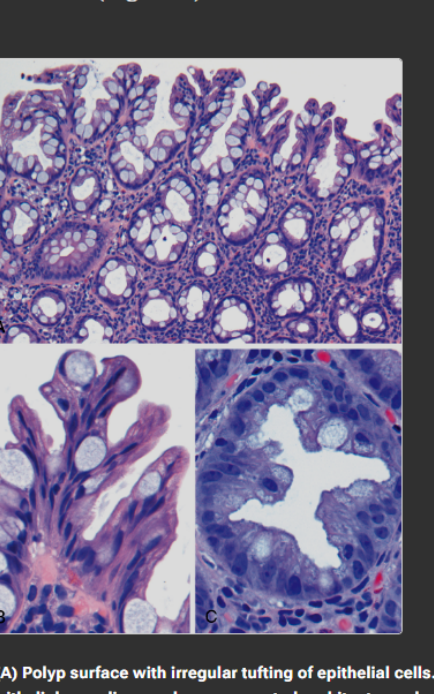


Figure 2. Hyperplastic polyp. (A) Polyp surface with irregular tufting of epithelial cells. (B) Tufting results from epithelial overcrowding. (C) Epithelial crowding produces a serrated architecture when crypts are cut in cross section.

CLINICAL CORRELATION

Finding a hyperplastic the frequency of subsequent colonoscopies.

Inflammatory Polyps

Inflammatory polyps usually form due to repeated cycles of injury and healing. An example would be inflammatory pseudopolyps that form in inflammatory bowel disease due to mucosal erosion.

Hamartomatous Polyps

Hamartomatous polyps can occur sporadically but are necessary to learn because they can be components of genetically determined or acquired syndromes. The main syndromes that are associated with hamartomatous polyps are Peutz-Jeghers syndrome and juvenile polyposis.

Peutz-Jeghers syndrome is a rare autosomal dominant condition, caused by mutations in *STK11*, that presents in childhood (median age 11 years) with multiple GI hamartomatous polyps (small intestine>colon>stomach) (Figure 3) and mucocutaneous hyperpigmentation.

CLINICAL CORRELATION

The hyperpigmentation is dark blue to brown macules on the lips, nostrils, buccal mucosa, palmar surfaces of the hands, genitalia, and perianal region.

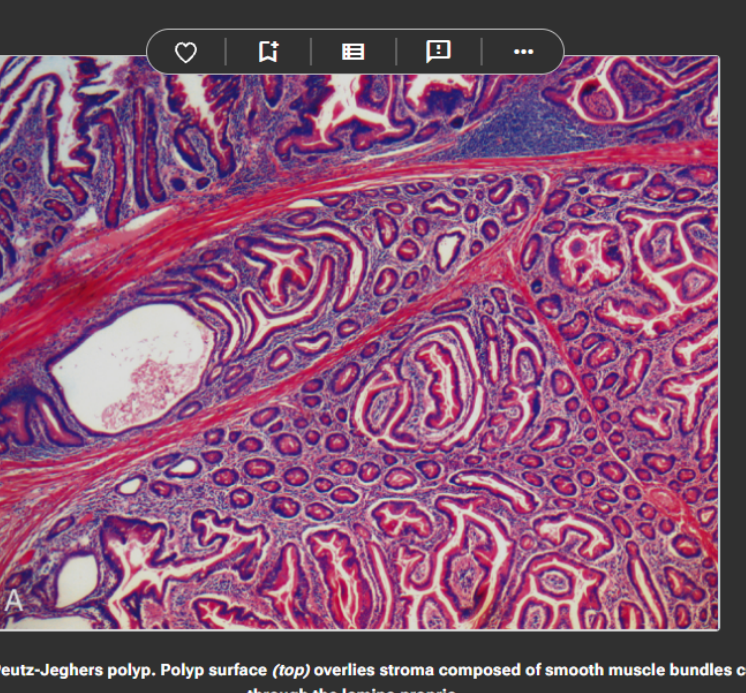


Figure 3. Peutz-Jeghers polyp. Polyp surface *(top)* overlies stroma composed of smooth muscle bundles cutting through the lamina propria.

Importantly, Peutz-Jeghers syndrome carries a high malignancy risk (~40%), requiring lifelong surveillance starting in childhood.

Juvenile polyposis syndrome is an autosomal dominant condition, with mutations in *SMAD4*, characterized by a few to hundreds of hamartomatous polyps. These patients have an increased risk of gastric, small intestinal, colonic, and pancreatic adenocarcinoma. Additionally, they have various congenital malformations such as pulmonary arteriovenous malformations.

CLINICAL CORRELATION

Most patients with juvenile polyposis syndrome are symptomatic by age 20 years, presenting with rectal bleeding, anemia, or obstruction; it may also cause diarrhea or rectal prolapse.

Neoplastic Polyps

Adenomatous polyps are the most prevalent neoplastic polyps in the colon; they comprise two-thirds of all colonic polyps. Increasing age and body mass index are risk factors for developing adenomatous polyps. Most of the time, these polyps do not cause symptoms and are detected on colon cancer screening.

Adenomatous polyps usually grow slowly (an average of 0.5 mm/year), and they do not commonly regress. The ones that are more likely to progress have certain features, as discussed below. Adenomatous polyps rarely progress to cancer, with only 5% of cases becoming malignant.

Adenomatous polyps can be classified by a number of characteristics, including endoscopic features, size, histology, and degree of dysplasia. Dysplasia refers to disordered, precancerous epithelial cell growth.

On endoscopy, adenomatous polyps can be:

- Sessile: the base and top of the lesion have the same diameter
- Pedunculated: the base has a smaller diameter than the top
- Flat or depressed

Histologically, adenomatous polyps can be tubular, villous, and tubulovillous. Figure 4 shows a microscopic comparison of normal colonic mucosa on the left and an adenomatous polyp on the right. Compared to normal, the neoplastic glands are more irregular with hyperchromatic, crowded nuclei and loss of goblet cells.

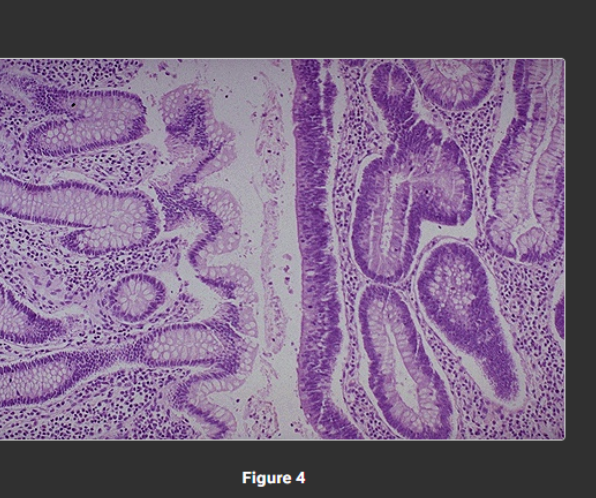


Figure 4

So which features convey a higher risk of malignancy?

- Polyps that have villous histology
- Sessile shape on endoscopy
- Adenomatous polyps >10 mm in size
- High-grade dysplasia in an adenomatous polyp
- Presence of three or more polyps

CLINICAL REASONING

Identifying adenomatous polyps is important, because it changes the frequency of colonoscopy cancer screening. Instead of every 10 years, the frequency can change to every 1-7 years.

Sessile serrated polyps, also referred to as sessile serrated adenomas, overlap histologically with hyperplastic polyps but are more commonly found in the right colon. Despite their malignant potential, these polyps lack typical cytologic features of dysplasia that are present in other adenomas. Histologically, these lesions can be differentiated from hyperplastic polyps by the presence of serrated architecture throughout the full length of the glands including the crypt base, crypt dilation, and lateral crypt extension.

What Are the Hereditary Colon Cancer Syndromes (LO #2)?

Hereditary cancer syndromes are inherited mutations that increase cancer risk. We're going to discuss the following notable hereditary colon cancer syndromes: familial adenomatous polyposis, Lynch syndrome, Gardner syndrome, and Turcot syndrome.

Familial adenomatous polyposis (FAP) is a syndrome that entails the presence of anywhere between 100 and 1000 colorectal adenomatous polyps (Figure 5). FAP is caus tumor suppressor gene *APC* (adenomatous po 5q21-22. FAP is inherited in an autosomal dominant pattern with near complete penetrance.

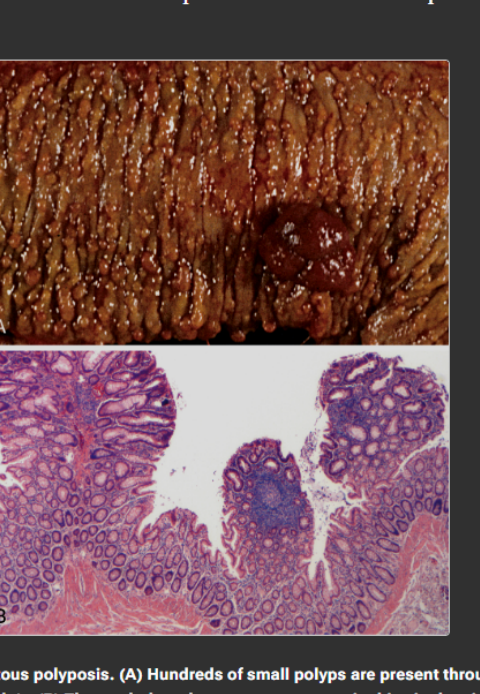


Figure 5. Familial adenomatous polyposis. (A) Hundreds of small polyps are present throughout this colon with a dominant polyp (right) . (B) Three tubular adenomas are present in this single microscopic field.

Most people with FAP are asymptomatic until they present with symptoms of colorectal cancer including abdominal pain, gastrointestinal (GI) bleeding, and diarrhea. Patients with FAP also have a high rate of upper GI tract polyps, typically fundic gland polyps and duodenal adenomas. The majority of patients are young at diagnosis rgo rigorous colorectal cancer scre

CLINICAL CORRELATION

Colorectal adenocarcinoma develops in 100% of untreated FAP patients, often before age 30 and nearly always by age 50.

Prophylactic colectomy may be necessary. Indications for colectomy include development of colorectal cancer, a marked increase in the number of polyps on consecutive exams, and symptoms related to colon cancer (ie, GI bleeding).

Lynch syndrome (formerly hereditary nonpolyposis colorectal cancer) is also an autosomal dominant disease caused by a germline mutation in DNA mismatch repair (MMR) genes that include *MLH1*, *MSH2*, *MSH6*, and *PMS2*. This is the most common inherited colorectal cancer.

Besides colorectal cancer, Lynch syndrome also increases the risk of other cancers, including endometrial, ovarian, and skin cancers (sebaceous neoplasms). Colorectal cancers in Lynch syndrome predominantly involve the right (ascending) colon.

Evaluation of both FAP and Lynch syndrome requires a thorough family history. There are guidelines for genetic testing that depend on what the family history yields.

What is the inheritance pattern of transmission for FAP and Lynch syndrome?

We should also briefly touch on two other conditions: Gardner syndrome and Turcot syndrome:

- **Gardner syndrome**, or familial colorectal polyposis, is a subtype of FAP that presents with polyps in the colon and cancers outside the colon including osteomas of the skull, thyroid cancer, epidermoid cysts, and fibromas.
- **Turcot syndrome**, which may be associated with FAP or Lynch syndrome, is a condition characterized by adenomatous colon polyps and an increased risk of colorectal cancer and brain cancer (i.e. medulloblastoma, glioma).

What are features of high-risk adenomatous polyps?

CLINICAL CORRELATION

***MUTYH*-associated polyposis** is an autosomal recessive polyposis syndrome caused by mutations in the *MUTYH* base excision repair gene. These individuals develop between 10 to 100 colorectal polyps (usually adenomatous polyps) and have an increased risk for colon cancer, necessitating more frequent colorectal cancer screening. Individuals with *MUTYH*-associated polyposis have an increased risk of gastric and duodenal polyps as well as osteomas, dental cysts, desmoids, sebaceous gland tumors, and thyroid cancer.

What Are Carcinoid Tumors (LO #3)?

Carcinoid tumors are a set of well-differentiated neuroendocrine tumors (NETs) that can originate in the digestive tract (55%), bronchopulmonary system (30%), and less commonly in the ovaries or kidneys. These tumors arise from enterochromaffin (EC) and enterochromaffin-like (ECL) cells.

CLINICAL CORRELATION

These tumors can also occur in the appendix.

Symptoms include flushing, diarrhea, and bronchospasm due to secretion of vasoactive substances (e.g., serotonin). Generally speaking, systemic symptoms such as skin flushing occur from carcinoid tumors of the small bowel (most commonly midgut) that metastasize to the liver.

Table 1

	Locations	Substances secreted	Potential for carcinoid syndrome
Tumors of foregut	Stomach Duodenum	Histamine 5-Hydroxytryptophan	Rare
Tumors of midgut	Jejunum Ileum Appendix Ascending colon	Kallikrein Prostaglandins Serotonin	Classic
Tumors of hindgut	Transverse colon Descending colon Sigmoid colon Rectum	Varies	Rare

When a carcinoid tumor exhibits systemic features caused by these vasoactive substances, it is termed **carcinoid syndrome**. Carcinoid syndrome causes episodic flushing, sec often triggered by stress, food, or alcohol.

CLINICAL CORRELATION

Carcinoid syndrome is associated with vitamin B3 (niacin) deficiency. Niacin deficiency can lead to pellagra, whose clinical hallmark is the triad of dermatitis, diarrhea, and dementia.

Treatment of these tumors usually involves resection, with some degree of post-operative surveillance.

What Is Colon Cancer (LO #3)?

Colon cancer or colorectal carcinoma (CRC) is the third leading cause of cancer-related death in both men and women. The number of patients with CRC diagnosed with screening methods has increased over the years.

Symptoms and Progression

The majority of patients with early-stage colon cancer have no symptoms; after symptoms begin, 70%-90% of CRCs are diagnosed. Typically, symptoms of CRC are due to growth of the tumor into the lumen or other nearby structures. This means that CRC diagnosed after symptom onset typically represents CRC at a relatively advanced stage.

The most common sy nges in bowel habits (74% of the time), rectal bleeding along with changed bowel movements (most common combination symptom), rectal mass, abdominal mass, and iron-deficiency anemia. Rarely, the only presenting symptom of CRC is abdominal pain.

CLINICAL REASONING

Remember from the blood and lymph course that iron-deficiency anemia in an adult male or post-menopausal female is colon cancer until proven otherwise!

Progression of colon cancer has been linked to a number of important genetic transformations. Two well-studied pathways are microsatellite instability pathway and the *APC/beta-catenin* pathway, which accounts for most sporadic CRCs.

The classic adenoma-carcinoma sequence (***APC/beta-catenin* pathway**) accounts for up to 80% of sporadic colon cancers and usually includes a mutation in *APC* early in the neoplastic process (Figure 6). Both copies of the *APC* gene must be functionally inactivated for adenomas to develop. As the process continues, additional mutations in *KRAS, SMAD2, SMAD4,* and *TP53* contribute to unrestrained cell growth and inhibition of apoptosis.

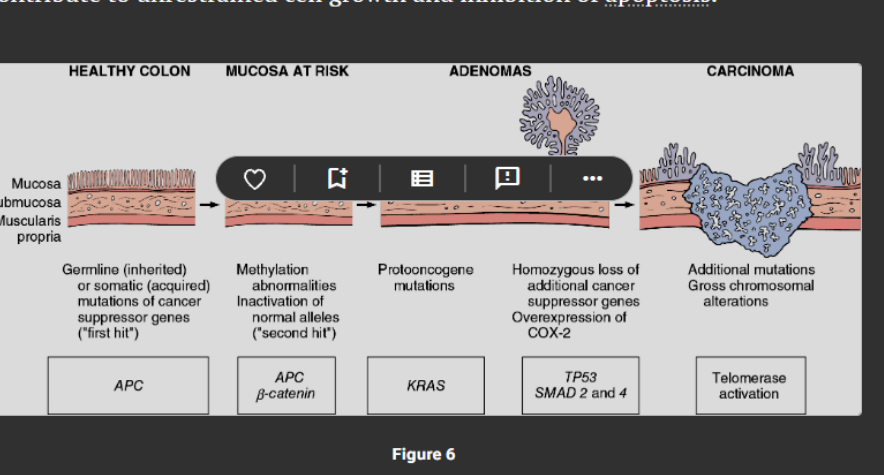

HEALTHY COLON
MUCOSA AT RISK
ADENOMAS
CARCINOMA
Mucosa
Submucosa
Muscularis propria
Germline (inherited) or somatic (acquired) mutations of cancer suppressor genes ("first hit")
Methylation abnormalities Inactivation of normal alleles ("second hit")
Protooncogene mutations
Homozygous loss of additional cancer suppressor genes Overexpression of COX-2
Additional mutations Gross chromosomal alterations
APC
APC β-catenin
KRAS
TP53 SMAD 2 and 4
Telomerase activation

Figure 6

The **microsatellite pathway** involves mutations or methylation in mismatch repair genes, which can cause some sporadic CRCs as well as Lynch syndrome (Figure 7).

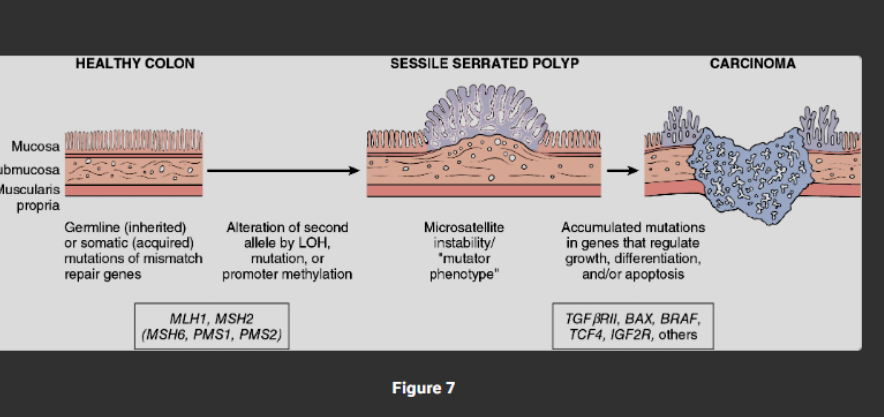

HEALTHY COLON
SESSILE SERRATED POLYP
CARCINOMA
Mucosa
Submucosa
Muscularis propria
Germline (inherited) or somatic (acquired) mutations of mismatch repair genes
Alteration of second allele by LOH, mutation, or promoter methylation
Microsatellite instability/ "mutator phenotype"
Accumulated mutations in genes that regulate growth, differentiation, and/or apoptosis
MLH1, MSH2 (MSH6, PMS1, PMS2)
TGFβRII, BAX, BRAF, TCF4, IGF2R, others

Figure 7

CLINICAL CORRELATION

Colon cancer involvi gh) pathway are more associated with sessile serrated lesions, prominent mucinous differentiation and peritumoral lymphoid infiltrates, and are frequently located in the right colon.

Risk Factors and Screening

The risk factors for CRC are varied and intertwined. Genetic predisposition plays a big role. Low socioeconomic status is associated with CRC, potentially based on modifiable factors such as suboptimal diet, lack of exercise, smoking, and obesity. Age is also a major risk factor for sporadic CRC. Though most CRC arises after age 50 years, recent data suggest that CRC diagnoses are increasing in those younger than 50 years and decreasing in those older than 50 years. Inflammatory bowel disease, abdominal radiation, diabetes and insulin resistance, alcohol, and processed meats also increase the risk of colorectal cancer.

For patients with CRC risk factors, screening is very important. Colorectal cancer screening typically begins at age 45 years for those at average risk. There are a variety of ways to screen, subdivided into tests that visualize the colon and tests that are stool-based. The choice is based on the test's advantages and disadvantages and patient preference.

These screening tests are discussed in greater detail below, but the recommendations typically consist of one of the following, depending on the patient:

- Colonoscopy every 10 years or FIT (fecal immunochemical test) every year
- CT colonography every 5 years, FIT-DNA every 3 years, or flexible sigmoidoscopy ev
- Capsule colonosc

Here is an overview of stool-based tests:

- **Guaiac-based fecal occult blood test (gFOBT):** This is best done as a take-home kit and entails testing several stool samples. This test has a low sensitivity for polyps and low specificity for disease. gFOBT has a high rate of false-positive results. Its advantage is in that it is non-invasive and inexpensive. It must be done annually. If gFOBT is positive, a timely colonoscopy is indicated.
- **Fecal immunochemical test (FIT):** This is more expensive than gFOBT but more convenient. It usually involves one stool sample. FIT uses specific antibodies to detect human hemoglobin in stool. Positive tests are followed by timely colonoscopy.
- **FIT-DNA:** This is a multitargeted stool DNA test with immunochemical assay and molecular markers for DNA mutations in the stool. This test also involves a single stool sample.

Tests that visualize the colon include endoscopy and radiography:

- **Colonoscopy** has high sensitivity (ability to catch disease) and specificity (ability to identify disease) for colon cancer. It also offers diagnostic and therapeutic advantages in that lesions can be biopsied during the procedure and sent for pathologic evaluation. All polyps are removed. The downside to a colonoscopy is the bowel prep: drinking almost a gallon of unappetizing liquid that results in profuse stooling. Another drawback is the need for sedation, which carries its own risk. The procedure also carries a risk of bowel perforation or bleeding.
- **CT colonography** is nearly as sensitive as colonoscopy. It also requires vigorous bowel prep but does not offer the therapeutic interventions made possible by color ned on samples. CT colonography must be done more frequently than colonoscopy for adequate screening. Repeated tests may increase cancer risk by cumulative doses of radiation.
- **Flexible sigmoidoscopy** only identifies lesions in the distal 60 cm of the colon but can be performed with minimal bowel prep. Abnormal findings may necessitate colonoscopy to visualize the entire colon.
- **Capsule colonoscopy** is a technique where the patient swallows a double-ended capsule that contains a small video device. This procedure also does not require sedation but does require bowel prep. Detected lesions necessitate a colonoscopy.

What is the most sensitive screening test for CRC?

Morphology

Though colon cancer , there are some general associations: tumors in the proximal colon usually grow as polypoid, exophytic masses that rarely cause obstruction, while carcinomas in the distal colon tend to be annular lesions that produce constrictions and luminal narrowing (Figure 8).

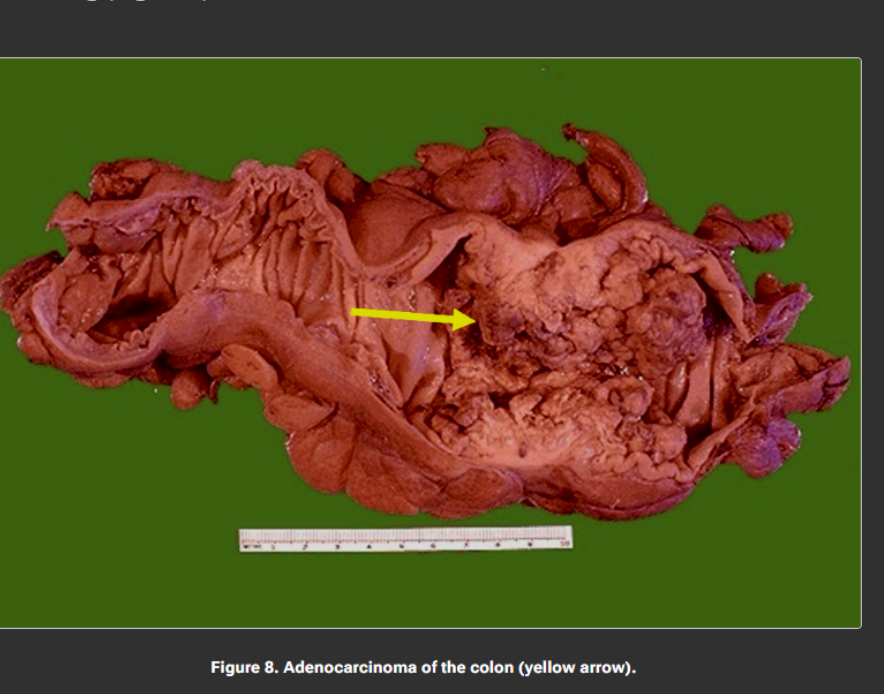


Figure 8. Adenocarcinoma of the colon (yellow arrow).

Microscopically, the majority of colon cancers are adenocarcinomas (producing glands) (Figure 9). The two most important prognostic factors for colorectal adenocarcinoma are depth of invasion and presence of lymph node metastases.

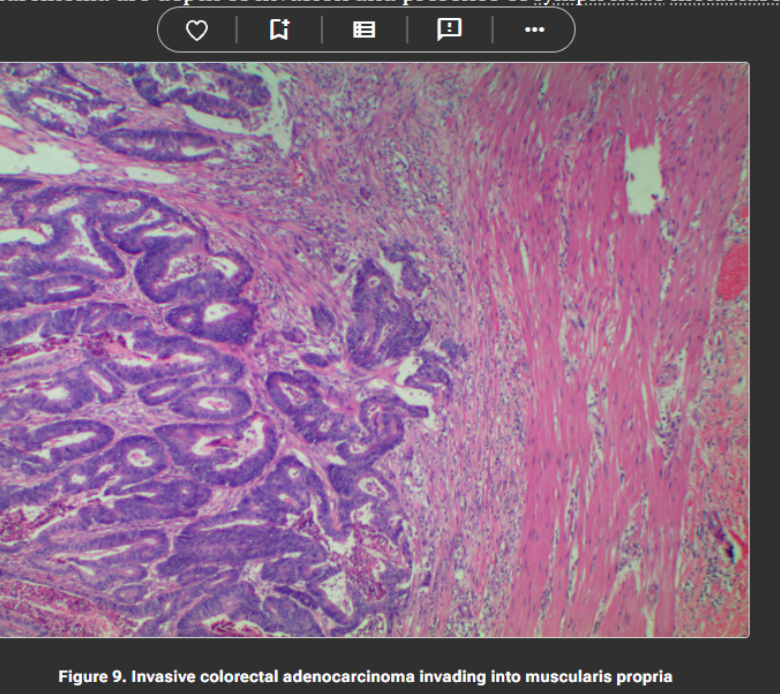


Figure 9. Invasive colorectal adenocarcinoma invading into muscularis propria

Management

Once a patient has been diagnosed with invasive colorectal adenocarcinoma, optimal management includes pretreatment clinical staging by physical examination, CT scan of the abdomen and pelvis, and chest imaging.

The main tumor marker that has been associated with colon cancer is **carcinoembryonic antigen (CEA)**. Serum CEA should **not** be used as a screening test for colon cancer due to low sensitivity and specificity for early stage disease; CEA is primarily obtained preoperatively to see how it normalizes following

As 80% of colon cancers are localized to the colon wall and/or regional lymph nodes, surgical resection is the only curative modality. Patients are followed up with adjuvant systemic therapy as clinically indicated. Patients with microsatellite instability tumors may benefit from neoadjuvant immunotherapy as these tumors respond to immune checkpoint inhibitors.

Carcinoid tumors are managed with surgical resection if localized and somatostatin analogs for symptomatic or metastatic disease.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to SC, how do you explain why you think a screening colonoscopy is important?

You understand that screening decisions are shared decisions between doctor and patient, and you want to provide helpful information on colon cancer so SC can make the best decision. You explain that colon cancer is the third most common cancer in the United States. It starts in small polyps that generally do not cause symptoms. By the time symptoms appear, the cancer may be far advanced. The purpose of the colonoscopy is to identify and remove the polyps before the cancer has time to appear. You also explain that he is at increased risk because of his age and family history. "Okay, doc, let me think about it." You give SC the brochure on colon cancer screening. The next day, he calls the office to let you know he is ready to schedule his colonoscopy. "I read about those polyps, and I read about colon cancer. Let's just say it's something I don't want," he tells you.

Summary

- Intestinal polyps are mucosal protuberances classified as non-neoplastic (hyperplastic, inflammatory, hamartomatous) or neoplastic (adenomatous, sessile serrated).
- Hamartomatous polyposis syndromes such as Peutz-Jeghers syndrome (STK11 mutation) and juvenile polyposis syndrome (SMAD4 mutation) are autosomal dominant conditions associated with GI polyps and significantly increased risks of multiple cancers.
- Familial adenomatous polyposis (FAP), caused by germline APC mutations, leads to hundreds of colorectal adenomatous polyps and progresses to colorectal adenocarcinoma in 100% of untreated patients; Lynch syndrome, caused by DNA mismatch repair gene mutations, is the most common inherited colorectal cancer syndrome; both are inherited in an autosomal dominant fashion.
- Carcinoid tumors are well-differentiated neuroendocrine tumors most commonly arising in the digestive tract; when they metastasize to the liver, systemic secretion of vasoactive substances such as serotonin causes carcinoid syndrome, characterized by episodic flushing, secretory diarrhea, and bronchospasm, and can lead to niacin deficiency and pellagra.
- Colorectal carcinoma (CRC) is the third leading cause of cancer-related death and commonly presents with changes in bowel habits, rectal bleeding, and iron-deficiency anemia; progression is driven by the APC/beta-catenin chromosomal instability pathway (80% of sporadic cases) or the microsatellite instability pathway, the latter being associated with right-sided tumors and responsiveness to immune checkpoint inhibitors.
- CRC screening begins at age 45 for average-risk individuals and includes colonoscopy ever n, with alternatives such as annual FIT, FIT-DNA every 3 years, CT colonography every 5 years, and capsule colonoscopy; surgical resection remains the only curative modality, with CEA used postoperatively to monitor treatment response rather than as a screening tool.

Review Questions

1. A 50-year-old male comes to the physician for routine screening. You alert him to the need for colorectal cancer screening. Which of the following tests is the most sensitive and specific for CRC screening?

A. ○ Colonoscopy
B. ○ CT colonography
C. ○ Flexible sigmoidoscopy
D. ○ Guaiac-based fecal occult blood test
E. ○ Kidney, ureter, and bladder (KUB) imaging

Explanation (requires correct answer)

2. A 21-year-old male comes to the physician due to abdominal pain and blood in his stool. The abdominal pain has been present for the last 5 months, and the bl st few weeks. He has no past medic n cancer at 38 years of age. A colonoscopy reveals more than 100 colonic polyps. Biopsy results of one of the polyps are consistent with colorectal carcinoma. Which of the following is the next most appropriate step in management?

A. ○ Colectomy
B. ○ Observation with serial digital rectal exams
C. ○ Reassurance
D. ○ Repeat colonoscopy in 5 years
E. ○ Repeat colonoscopy in 6 months

Explanation (requires correct answer)

3. A 54-year-old man undergoes a screening colonoscopy, which reveals a 14-mm sessile polyp in the left colon with villous histology and moderate dysplasia. He is asymptomatic, and his past medical history is unremarkable except for obesity. The polyp is completely resected and sent for histological evaluation. He asks his physician about the likelihood that his polyp will turn into cancer and how often he should be screened. Which feature of this polyp is most strongly associated with increased risk of malignant transformation?

A. ○ Peduncul
B. ○ Presence of a single polyp
C. ○ Tubular histology
D. ○ Villous histology

Explanation (requires correct answer)

4. A 16-year-old boy presents with rectal bleeding and anemia. He notes dark red blood mixed with stool and frequent, painless bowel movements. Examination reveals pale conjunctiva, mild abdominal tenderness, and soft masses prolapsing from the rectum. Family history is notable for his father having surgery for intestinal polyps at a young age. Colonoscopy reveals numerous hamartomatous polyps throughout the colon and rectum, and genetic testing identifies a mutation in the *SMAD4* gene. Which additional clinical finding is most likely in this patient based on his diagnosis?

A. ○ Cerebral medulloblastoma
B. ○ Mucocutaneous hyperpigmentation
C. ○ Osteomas of the skull
D. ○ Pulmonary arteriovenous malformations
E. ○ Sebaceous skin neoplasms

E. ○ Sebaceous skin neoplasms

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Compare and contrast the histologic types of intestinal polyps, including their genetic associations and clinical significance.
02 ☐ Describe the genetics and malignancies associated with the included hereditary colon cancer syndromes.
03 ☐ Differentiate the clinical presentation, pathophysiology, diagnosis, and management of colorectal carcinoma and carcinoid tumor.

Objective Confidence: 0/3